* SIFT Descriptor

→ Given a Keypoint, SIFT detector works as follows:

① Grid Setup

→ Take a 16 × 16 patch around the Keypoint

→ Divide it into 16 regions (4 × 4 grid).

→ Each region is 4 × 4 pixels

② Orientation Histograms:-

→ For each region, compute a histogram of gradient directions

→ Use 8 bins (each covering 45°).

→ Apply a Gausian weight centered on the Keypoint, with.

$$\sigma = \frac{1}{2} \times \text{grid size} = 8$$

③ Concatenate Histogram:

→ Each region → 8 values

→ 16 regions × 8 → 128 dimensional descriptor

④ Normalize the descriptor to unit length.

→ Pixels larger than 0.2 are saturated and normalized again.

16×16 → 4 → 8 bins each histogram apply gaus

$\sigma = \frac{1}{2} \times$ grid size $= 8$ — concatenate histo gram

16 × 8 = 128 dimension

4 → normalize desc to unit length length 0.2

* Trilinear interpolation

→ Bins are assigned in a soft manner to avoid boundary effects.

→ The contribution of a pixel is spread between its two adjacent bins weighted by the distance to the bin centers:

↓

Distribution contribution b/w :-

→ two adjacent bins (direction-wise)

→ four neighboring regions (position wise)

→ Weighting :-

$$weight_K = 1 - d_K \quad , \quad weight_{K+1} = 1 - d_{K+1}$$

* SIFT is scale, rotation & affine intensity change invariant.

* Keypoint Matching

→ Match Keypoints from target image (T) to reference image (R)

→ Solve a NN problem using Euclidean distance b/w descriptors.

Matching Criteria

① Thresholding

→ Accept the match only if:

$$d_{NN} \leq T$$

② Ratio Test (to avoid ambiguity)

→ let :-

- d_{NN} : Distance to closest match.
- d_{NN2} : Distance to second-closest match.

→ Accept only if:

$$\frac{d_{NN}}{d_{NN2}} \leq T$$

→ Prevents confusion when two matches are close together.

* Efficient Nearest Neighbor Search

→ Exhaustive search is too slow, so we:

↳ Indexing technique:

1. K-d tree
2. With Best Bin First (BBF) strategy
   - ↳ Not optimal but fast
   - ↳ Prioritize the most likely bin first, then expands if needed.